Spatial–Spectral Transformer With Cross-Attention for Hyperspectral Image Classification

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***Abstract*— Convolutional neural networks (CNNs) have been widely used in hyperspectral image (HSI) classification tasks because of their excellent local spatial feature extraction capabilities. However, because it is difficult to establish dependencies between long sequences of data for CNNs, there are limitations in the process of processing hyperspectral spectral sequence features. To overcome these limitations, inspired by the Transformer model, a spatial–spectral transformer with cross-attention (CASST) method is proposed. Overall, the method consists of a dual-branch structures, i.e., spatial and spectral sequence branches. The former is used to capture fine-grained spatial information of HSI, and the latter is adopted to extract the spectral features and establish interdependencies between spectral sequences. Specifically, to enhance the consistency among features and relieve computational burden, we design a spatial–spectral cross-attention module with weighted sharing to extract the interactive spatial–spectral fusion feature intra Transformer block, while also developing a spatial–spectral weighted sharing mechanism to capture the robust semantic feature inter Transformer block. Performance evaluation experiments are conducted on three hyperspectral classification datasets, demonstrating that the CASST method achieves better accuracy than the state-of-the-art Transformer classification models and mainstream classification networks.**

***Index Terms*— Convolutional neural network (CNN), cross-attention, hyperspectral image (HSI) classification, local spatial features, long sequence data, transformer.**

I. Introduction

HYPERSPECTRAL image (HSI) data are acquired by imaging spectrometers, which can provide a large amount of narrow-band spectral information from the visible spectrum to the infrared spectrum for each pixel [1], and generate a complete and continuous spectral curve. HSI data not only has rich spectral band information but also contain the spatial features of ground cover. Therefore, compared with RGB images and multispectral images, HSI data have better object representation ability and more accurate feature identification ability [2]. With the development of remote sensing imaging technology, HSI plays an important role in various remote sensing fields and is widely used in various scenarios such as mineral exploration [3], army research [4], and precision agriculture [5]. HSI classification is usually a pixel-level classification task, i.e., it involves obtaining pixel-level labels of the entire image by learning prior knowledge [6], [7], [8].

Researchers have made many attempts to achieve accurate land object classification. Farid and Bruzzone [9] effectively completed the classification task by introducing the support vector machine (SVM) classifier into HSIs. Wang *et al*. [10] applied a morphological algorithm to the hyperspectral classification task and used extended multiattribute profiles (EMAPs) for dimension-reduced HSI data to fully mine its spatial structural features under diverse attributes. In [11], The geodesic-based sparse manifold hypergraph (GSMH) method is proposed to reveal the complex manifold structure and multivariate relationship of samples in HSI and to solve the problem of weak classification performance of the traditional sparse representation (SR) algorithms for small size training samples. However, for the feature extraction process of traditional algorithms, when the number of training samples gradually increases, there is a certain performance bottleneck, which leads to incomplete utilization of HSI data.

In recent years, algorithms based on deep learning have brought new inspirations to the field of remote sensing applications [12], [13], [14] and have made further breakthroughs in the HSI classification task [15], [16], [17]. Its adaptive feature capture mechanism effectively overcomes the drawbacks of the traditional machine learning algorithms and improves feature identification. Chen *et al*. [18] designed a stacked autoencoder (SAE) structure model for the HSI classification task, enabling more adequate feature extraction. Zhong *et al*. [19] improved the deep belief network (DBN) to improve the representation ability of HSI features. Zhu *et al*. [20] applied generative adversarial networks (GANs) to the HSI classification task, which alleviated the overfitting problem in convolutional

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neural networks (CNNs) and noticeably improved the classification accuracy.

Recurrent neural network (RNN) [21] is widely used in the field of natural language processing (NLP) because it is very effective in processing data with sequence features and can build the semantic information of long sequence features. In addition, considering that the spectral features of HSI have sequential attributes, RNNs are used as a spectral feature extraction tool and can be effectively transferred to the HSI classification task. Mou *et al*. [22] used the RNN model to perform pixel-level processing of HSI data from sequential perspectives to establish correlations between spectral sequence features. Zhang *et al*. [23] proposed a local spatial sequential method, which initially captured the spatial–spectral features of HSI data and implanted the RNN model to further obtain high-level semantic features. Atharva *et al*. [24] proposed a patch-based RNN model with long short term memory (LSTM) units, which can simultaneously process multitemporal, multispectral, and multispatial features of remote sensing data. It is worth noting that although RNNs or LSTM can effectively characterize the spectral sequence features, they find it difficult to fully use the spatial feature information of HSI.

As an excellent local neighborhood information capture tool, CNNs have been widely used in the field of remote sensing [25] and have a very attractive classification effect [26], [27], [28]. Ying *et al*. [29] proposed a CNN model based on 3-D convolution kernels to extract the joint spatial–spectral features of HSI. To process HSI data, Paoletti *et al*. [30] designed a new CNN model based on the spatial–spectral capsule networks to enrich network features and reduce the complexity of the framework. To capture finer spatial–spectral features, many hybrid network structures have been developed to combine the advantages of each network. Yang *et al*. [31] designed a dual-branch network structure combining 1-D CNN and 2-D CNN for extracting spectral and spatial features of HSI, respectively. In [32], an attention-based hybrid network structure of RNN and CNN is proposed to learn the spectral similarity of continuous bands and the spatial similarity of local neighborhood pixels. Hong *et al*. [33] proposed a novel miniGCN to overcome the gradient explosion or gradient vanishing problem and combined it with a CNN to enrich the diversity of features. However, owing to the size of its inherent convolution kernel, the receptive field in the shallow layer of the network is local, and it only has the ability to extract features at the global level in the deeper layers. In addition, HSI spectral features have sequential attributes, CNN models based on spatial context information capture have limitations in processing spectral features, and it is difficult to establish long-distance dependencies.

In recent times, Transformer models have shown considerable potential in the field of computer vision (CV) [34], [35], [36], [37]. By combining the residual-structure-based multihead self-attention module and multilayer perceptron (MLP), the model achieves performance comparable to or even surpassing the CNN model in tasks such as image classification, target detection, and target tracking. As Transformer have powerful spatial and long-distance sequence feature processing capabilities, researchers have tried to explore the prospects of this model in remote sensing fields and introduce it into the task of hyperspectral classification [38]. To effectively capture the spatial features of HSI and establish spectral sequence spectral relationships, He *et al*. [39] combined a pretrained VGG-like model with Transformer to overcome the drawbacks of a single CNN model in processing long sequence features. Qing *et al*. [40] introduced the spectral attention module before Transformer and proposed the self-attention-based transformer (SAT) network to enhance the feature extraction ability of the model. Hong *et al*. [41] developed the pure Transformer into the HSI classification task for the first time. By combining local bands to form group-wise spectral embeddings as the input of the Transformer, a classification process with the ability to represent local spectral sequence features was realized. Sun *et al*. [42] combined the CNN hybrid model and Gaussian weighted feature tokenizer with the Transformer to achieve feature preextraction while enhancing the separability of samples. However, the above methods either rely on the embedded CNN to realize the joint extraction process of spatial–spectral features or lack the complete capture of HSI spatial feature in the Transformer and ignore the effectiveness of its spatial context information. These insufficient sample information utilization process leads to the bottleneck of model performance.

Considering the efficient utilization of HSI spatial and spectral information by the dual-branch model, we improve the ViT structure and propose a novel classification model called the spatial–spectral transformer with cross-attention (CASST), which captures the spatial–spectral features, and at the same time, both the information are fully integrated to realize the HSI classification process with efficient feature utilization. Specifically, a dual-branch Transformer model is designed to efficiently extract the spatial and spectral sequence features of HSI. The convolution module is first used to realize the feature tokenization of the spatial and spectral branches of the network. In the process of transformer encoder (TE), an improved cross-attention module intra Transformer block is introduced to perform interactive feature fusion extraction between the spatial and spectral branches. After passing through the spatial–spectral weighted sharing MLP inter Transformer block, the two branch features will concatenate together and send to MLP head to implement robust classification.

1) The vanilla vision transformer (ViT) model is improved, and a new dual-branch Transformer consisting of spatial and spectral sequence branches is proposed to implement the HSI classification task. The former is used to capture fine-grained spatial information of HSI, and the latter is adopted to extract the spectral features and establish interdependencies between spectral sequences.
2) We design a spatial–spectral cross-attention (SSCA) module intra Transformer block to extract the interactive spatial–spectral fusion feature and enhance the sufficiency of information utilization.
3) We develop a spatial–spectral weighted sharing mechanism inter Transformer block to relieve computational burden and enhance the consistency among features.

The rest of this article is organized as follows. Section II describes the proposed CASST method. Section III presents

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an experimental analysis of CASST, including the experiment setup and comparison with various state-of-the-art methods on the Indian Pines, Pavia University, and Salinas datasets. Conclusions and future work are discussed in Section IV.

II. Description of the Proposed Approach

In this section, we first introduce the application of a Transformer structure in the CV domain. On this basis, a spatial–spectral Transformer structure based on cross-attention (CASST) is proposed to realize the application of the Transformer structure in hyperspectral classification scenarios. In addition, the function and importance of SSCA, weight-shared Transformer block, and the dual-branch model used in CASST are analyzed and explained in detail.

A. Description of ViT

The original purpose of a Transformer [43], [44] was to solve tasks in the field of NLP such as machine translation. The use of the self-attention mechanism overcomes the issues associated with the RNN model having a fixed sequence attribute, realizes parallel computing of long-sequence data, and imparts a stronger ability to capture long-distance features than the CNN model. The proposal of this method substantially promotes the development of the time-series data models. Recently, the application of a Transformer structure in the field of CV, ViT [45], has achieved excellent results in the image classification tasks, and to a certain extent has surpassed the state-of-the-art CNN models. ViT is the first attempt of the Transformer model in the CV field and fully demonstrates the feasibility of the Transformer in the image classification tasks.

Specifically, for the image $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, it is first transformed into a set of flat 2-D sequence blocks $\mathbf{x}_p \in \mathbb{R}^{N\times(P^2\times C)}$, where H, W, and C are the height, width, and channel number of the original image, respectively, P^2 represents the spatial scale of the sequence block, and N represents the number of sequence blocks, whose value is HW/P^2. Then, through a linear projection, the original sequence is mapped to the D dimension to form a new set of sequence vectors, which are called patch embeddings. After adding a learnable embedding vector $\mathbf{x}_{\text{class}}$ and a set of position embeddings $\mathbf{E}_{\text{pos}}$, the input of the TE, i.e., $\mathbf{z}_0$ is formed, and its expression is as follows:

$$\begin{aligned}\mathbf{x}_{\text{class}} &= W_0 \in \mathbb{R}^{1\times D}, \quad \mathbf{E}_{\text{pos}} = W_1 \in \mathbb{R}^{(N+1)\times D}\\ \mathbf{z}_0 &= \left[\mathbf{x}_{\text{class}}; \mathbf{x}_p^1\mathbf{E}; \mathbf{x}_p^2\mathbf{E}; \ldots; \mathbf{x}_p^N\mathbf{E}\right] + \mathbf{E}_{\text{pos}}\end{aligned} \tag{1}$$

where $\mathbf{x}_p^i$ represents the ith sequence block, and $\mathbf{E}$ is the linear projection process. It is worth noting that the learnable embedding vector $\mathbf{x}_{\text{class}}$ is attached to patch embeddings and regarded as a part of the feature encoding sequence. With the process of self-attention, $\mathbf{x}_{\text{class}}$ is trained together with another tokens and gradually receives image feature information and context feature dependencies. Position embeddings are used to store the position information of patch embeddings and play a crucial role in the Transformer structure, mainly because the self-attention mechanism in the Transformer lacks the ability to capture the order of sequence data.

The main reason this structure has high potential is the use of the multihead attention mechanism, which can easily realize long-distance feature correlation calculation and establish feature dependencies of global data. Fig. 1 shows the framework of the multihead attention structure. Specifically, for each head, three learnable weight matrices W_q, W_k, and W_v are first used to perform matrix multiplication with the Transformer input sequence to generate three new sequence vectors, namely, $Q \in \mathbb{R}^{(N+1)\cdot D/n}$, $K \in \mathbb{R}^{(N+1)\cdot D/n}$, and $V \in \mathbb{R}^{(N+1)\cdot D/n}$, where n represents the number of heads. Then, the dot product of each Q vector and K vector is calculated to form a dot product matrix. After the matrix is normalized, it is passed through the SoftMax layer to generate the attention weight score matrix of the sequence vectors. Finally, multiplying the score matrix by V completes the self-attention. The specific self-attention calculation process is as follows:

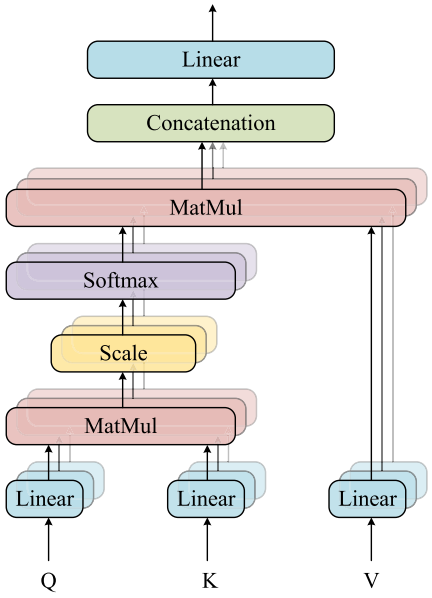


Fig. 1. Structure of multihead self-attention

$$\begin{aligned}Q_i &= \mathbf{z}_0 W_q, \quad K_i = \mathbf{z}_0 W_k, \quad V_i = \mathbf{z}_0 W_v\\ \boldsymbol{A}_i &= \text{softmax}\left(\frac{Q_i {K_i}^T}{\sqrt{d_k}}\right)\\ \mathbf{z}_i &= \text{Attention}(Q_i, K_i, V_i) = \boldsymbol{A}_i V_i\end{aligned} \tag{2}$$

where $\boldsymbol{A}_i$ and $\mathbf{z}_i$ are the attention map and the output of the ith head in self-attention, respectively, and d_k represents the dimension of the vector K. Owing to the existence of multiple heads, the heads of each head will be concatenated together to realize the calculation of multihead attention. The specific calculation process is as follows:

$$\text{MultiHead}(Q, K, V) = \text{concat}(\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n) W_O \tag{3}$$

where W_O is a linear transformation matrix, which aims to align the feature dimension between the output and the input of multihead attention. After multihead attention, the features with global dependencies can be obtained, and then an MLP is connected to complete the entire Transformer block process. The structure of MLP contains two fully connected layers and embeds a Gaussian error linear unit (GELU) activation function and a layer normalization (LN). It is worth noting that TE is composed of multiple Transformer blocks, which

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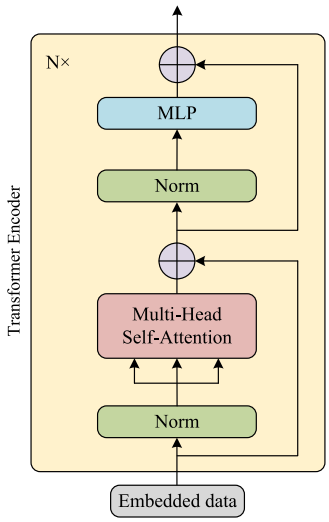


Fig. 2. Structure of TE

together complete the effective extraction of global features. The structure of TE is shown in Fig. 2

Replacing the patch embeddings, $\mathbf{x}_{\text{class}}$ does not carry obvious semantic information, so in the classification module of the network, $\mathbf{x}_{\text{class}}$ will replace the patch tokens to pass through the MLP head and be used as the final sample category representation, which is quite different from CNN.

B. Dual-Branch Transformer

Owing to the use of self-attention, the Transformer structure can not only enable the acquisition of adaptive spatial information but also establish the interdependence between long sequences. For the classification of hyperspectral data, the network structure of the dual-branch Transformer is proposed. The framework of CASST is shown in Fig. 3, where the upper is the spatial sequence branch, which is used for the extraction of fine-grained HSI spatial features. The lower is the spectral sequence branch, which effectively mines the spectral sequence information of HSI.

Specifically, for a sample pixel $\mathbf{x} \in \mathbb{R}^{1\times1\times C}$ in HSI, we initially expand it in the neighborhood to form a sample cube $\mathbf{p} \in \mathbb{R}^{k\times k\times C}$, where C is the number of bands in HSI, and k represents the spatial size of the cube. It should be mentioned that the segmentation of the image by ViT is coarse-grained, which ignores the details of the local space to a certain extent. Therefore, a 3×3 convolution kernel is implemented in the spatial division operation to obtain fine-grained blocks based on pixel level, which is to emphasize the importance of each block in the sample cube to the global context information as much as possible. Then, we divide the blocks in the spatial dimension and flatten each small block to form spatial sequence features. After each sequence vector passes through the linear function, the dimensions of them are converted into D, and the entire spatial tokenization process is completed. For the spectral sequence branch, HSI is initially grouped with the spectral dimension to form a spectral vector group $\mathbf{s} = \{\mathbf{s}_1, \mathbf{s}_2, \ldots, \mathbf{s}_m\} \in \mathbb{R}^{k\times k\times1}$, which is then passed through a simple CNN to further generate the spectral sequence features $\mathbf{y} = \{\mathbf{y}_1 \text{ and } \mathbf{y}_2, \ldots, \mathbf{y}_m\} \in \mathbb{R}^{1\times D}$, where m represents the number of spectral bands, and D represents the dimension of each spectral feature after tokenization. The structure of the small CNN model mainly includes a convolutional layer, a pooling layer, and a linear layer, which is to tokenize the features in HSI spectral dimension. The spectral tokenization is shown in Fig. 4.

After the dual-branch tokenization process, TE is used to effectively capture the spatial–spectral fusion features. It is worth noting that the inputs for different branches also contain different class tokens. After feature extraction of dual branch, the spatial class token and spectral class token are extracted separately, concatenated in channel dimension, and sent to the MLP head to obtain the classification category information pred. The specific process is as follows:

$$\begin{aligned} \mathbf{x}_{\text{class}} &= \text{Concat}\left(\mathbf{x}^{\text{spe}}_{\text{class}}\mathbf{x}^{\text{spa}}_{\text{class}}\right) \\ \text{pred} &= \text{Softmax}[f(\mathbf{x}_{\text{class}})] \end{aligned} \tag{4}$$

where $\mathbf{x}^{\text{spe}}_{\text{class}}$ represents the class token of the spectral branch, $\mathbf{x}^{\text{spa}}_{\text{class}}$ represents the class token of the spatial branch, and $f(\cdot)$ is the MLP head process.

C. Spatial–Spectral Cross-Attention (SSCA)

Cross-attention [46], [47] is a method for feature-level fusion in the Transformer, and it has been used in Transformer-based multimodal data fusion and multiscale feature fusion [48], [49]. To enhance the ability of information utilization, we have improved cross-attention and proposed SSCA module to extract the adaptive spatial and spectral fusion features intra Transformer block. SSCA replaces the single-end fusion method, realizes information interaction in feature coding, and ensures the sufficiency of feature fusion, which is shown in Fig. 5. Specifically, for the spatial branch sequence feature $\mathbf{x}^{\text{spa}}$ and spectral branch sequence feature $\mathbf{x}^{\text{spe}}$ after tokenization, the class token and feature embeddings of their respective branches are initially obtained. Then, the class token between the two branches is exchanged to concatenate it with the other branch and sent to the multihead self-attention module. The result is that the class token in one branch and the feature embeddings in the other branch perform self-attention computation, which realizes interactive learning between spatial and spectral features. After the calculation is completed, they will further add to their original class token, to implement the feature fusion between spatial and spectral features. During the SSCA process, the residual structure is also kept as the vanilla self-attention to avoid the problem of network degradation.

Taking the spatial branch sequence feature $\mathbf{x}^{\text{spa}}$ as an example, the cross-feature fusion process is shown as follows:

$$\begin{aligned} \mathbf{x}^{\text{spa}\ \prime}_{\text{class}} &= \text{Attention}\left[\text{Concat}\left(\mathbf{x}^{\text{spa}}_{\text{class}}, \mathbf{x}^{\text{spe}}_{\text{feature}}\right)\right] \\ \mathbf{x}^{\text{spa}\ \prime\prime}_{\text{class}} &= \mathbf{x}^{\text{spa}\ \prime}_{\text{class}} + \mathbf{x}^{\text{spa}}_{\text{class}} \\ \mathbf{x}^{\text{spa}\prime} &= \mathbf{x}^{\text{spa}} + \left[\text{Concat}\left(\mathbf{x}^{\text{spa}\ \prime\prime}_{\text{class}}, \mathbf{x}^{\text{spa}\ \prime}_{\text{feature}}\right)\right] \end{aligned} \tag{5}$$

where $\mathbf{x}^{\text{spa}}_{\text{class}}$ is the class token of the spatial branch, $\mathbf{x}^{\text{spe}}_{\text{feature}}$ is the feature embeddings of the spectral branch, Attention$(\cdot)$ is the multihead attention module, $\mathbf{x}^{\text{spa}\ \prime\prime}_{\text{class}}$ is the spatial class token after cross-feature fusion, and $\mathbf{x}^{\text{spa}\prime}$ is the output of SSCA.

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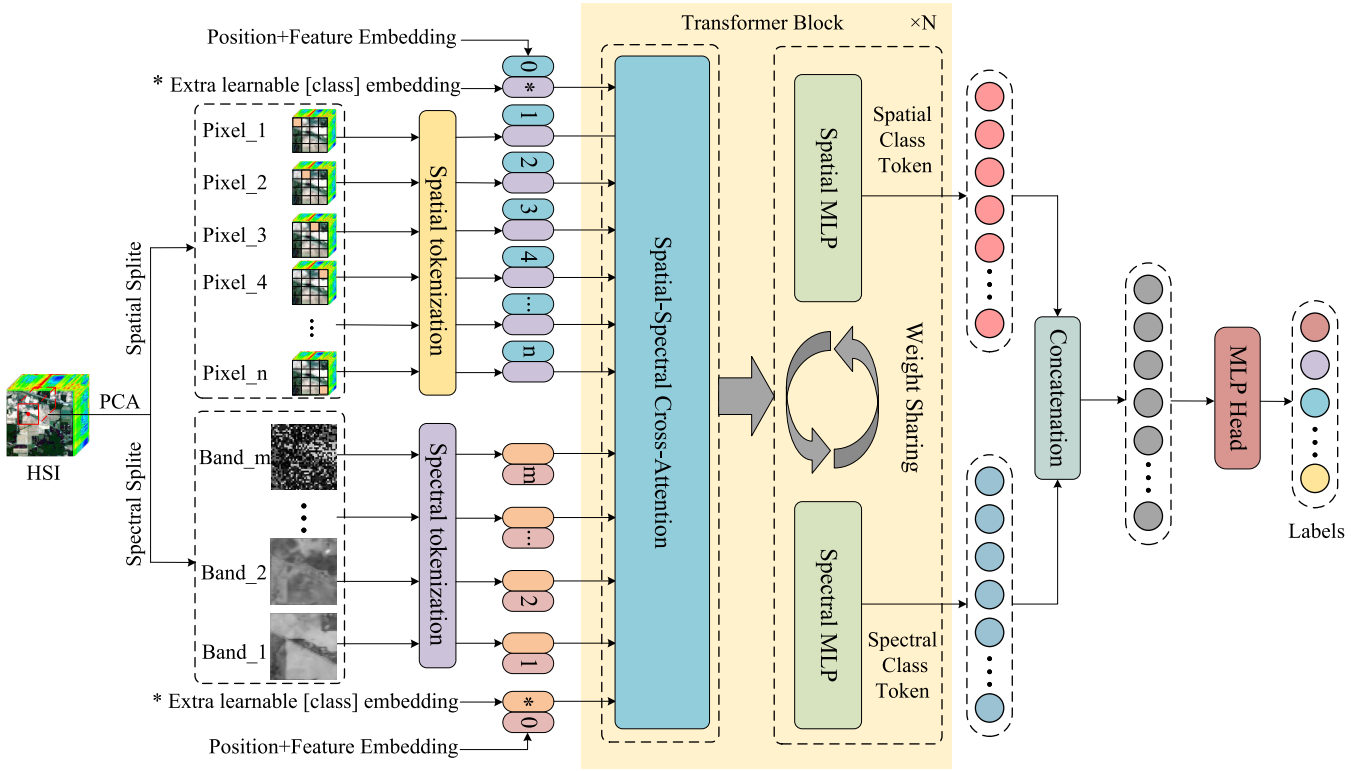


Fig. 3. Framework of the proposed CASST method, where the upper branch is the spatial sequence branch of the Transformer, and the lower is the respective spectral sequence branch.

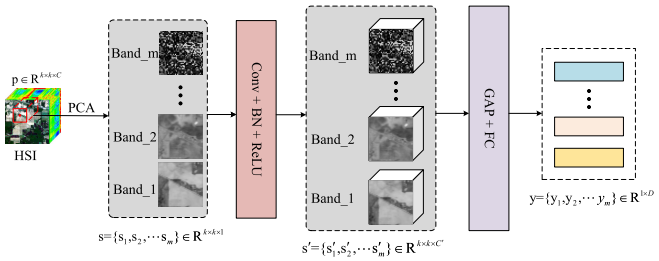


Fig. 4. Pross of spectral tokenization

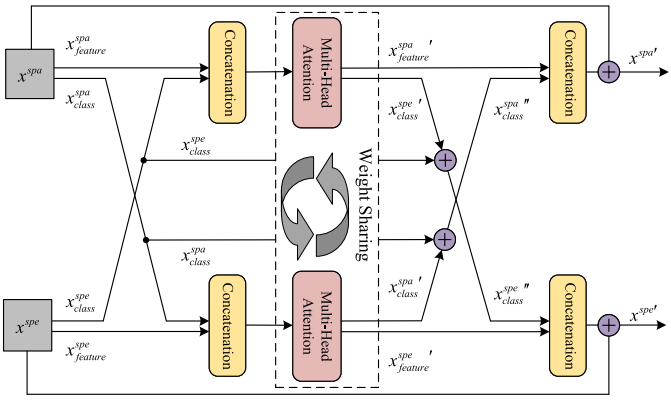


Fig. 5. Structure of SSCA

D. Weight-Shared Transformer Block

Since the self-attention in the Transformer has multiple matrix dot product operations, with the increase in feature embeddings, there is a substantial computational burden in self-attention, which is especially obvious in the multibranch Transformer structure. It is worth noting that the large number of linear layer operations in the Transformer block facilitates the sharing of weight parameters between the spatial and spectral branches. Specifically, we design a spatial–spectral weighted sharing mechanism consisting of multihead attention and MLP intra Transformer block, which relieves the pressure of network training and saves the training time. On the other hand, the weight sharing mechanism among the spatial and spectral sequence branches enhances the consistency between features and the robustness of the fusion process. The weighted sharing structure is shown in Figs. 5 and 3, respectively.

III. Experimental and Analysis

In this section, three well-known datasets (i.e., Indian Pines, Pavia University, and Salinas datasets) and three evaluation metrics [i.e., average accuracy (AA), overall accuracy (OA), and kappa coefficient (Kappa)] are applied to analyze the parameters of CASST and compare the performance of various state-of-the-art methods. All the algorithm implementation processes of the CASST method are completed on the python3.7 platform. The algorithm based on deep learning is implemented under the PyTorch framework. PyTorch is an open source machine learning framework proposed by Facebook, which not only provides a convenient deep learning system but also supports the code running process for graphic processing unit (GPU) acceleration. All our experiments are performed on a personal computer having a Windows 10 operation system, Intel Core i7-7800X central processing unit (CPU), 32-GB RAM, and an NVIDIA GeForce RTX 1080 Ti

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GPU. The implementation process of CASST is presented in Algorithm 1.

Algorithm 1 CASST

Input: The raw HSI data $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ and ground truth $\mathbf{Y} \in \mathbb{R}^{H\times W}$.

Output: Classification results of testing dataset and visualization map.

Reduce the dimension of HSI $\mathbf{X}$ and expand each pixel into spatial patch.
Obtain the training set, validation set, and testing set, then build their respective dataloaders
Set train batch size $b = 64$, optimizer stochastic gradient descent (SGD) with the learning rate $lr = 0.005$, and train epochs $e = 150$.
for $e = 1$ to 150 **do**
    Embedding the input of spatial patches along the spectral dimension through a simple CNN
    Embedding the input of spatial patches along the spatial dimension through an 3×3 convolution layer with the stride $s = 1$.
    Perform the SSCA module.
    Perform weighted sharing mechanism among spatial and spectral branches, respectively.
    Concatenate the class tokens of two branch.
    Classify the fused class token using the MLP head and SoftMax.
end for
Save the trained model to classify the testing set, and plot the visualization map.

A. Datasets

1) Indian Pines Dataset: The Indian Pines dataset was acquired by airborne visible infrared imaging spectrometer (AVIRIS) imaging an Indian pine tree in Indiana, USA. The spatial resolution of the resulting image is 20 m, and the spatial size is 145 × 145. In addition, the imaging wavelength range of the spectrometer is 0.4–2.5 μm, and 200 remaining bands are reserved for classification research after data processing. The Indian Pines dataset has a total of 10 249 ground objects labeled as per 16 test categories. Table I lists the number of samples in the training set, validation set, and testing set applied in the experiment. Fig. 6 presents the corresponding HSI pseudocolor image and total sample ground-truth map for this dataset.

2) Pavia University Dataset: The Pavia University dataset was collected by reflective optics system imaging spectrometer (ROSIS) sensors over Pavia, northern Italy. The spatial resolution of this dataset is 1.3 m, and the spatial size is 610 × 340. In addition, the collected bands ranged from 0.4 to 0.86 μm, and the remaining 103 bands were retained by removing the noise band. The Pavia University dataset has a total of 41 176 test sample points and nine test categories. Table II lists the number of samples in the training set, validation set, and testing set applied in the experiment, and Fig. 7 presents the corresponding HSI pseudocolor image and total sample ground truth map for this dataset.

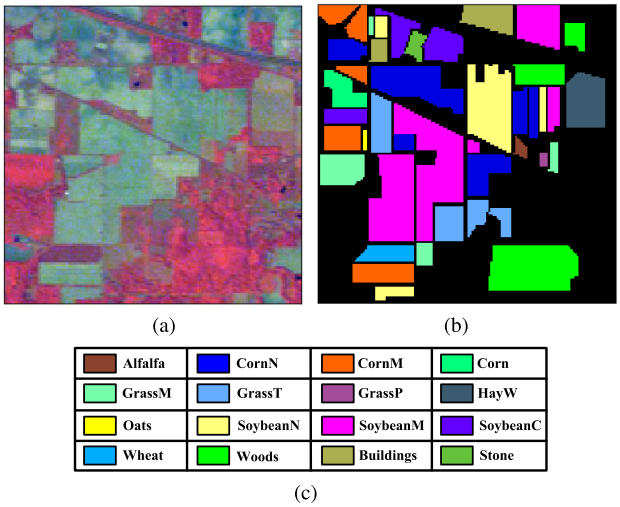


Fig. 6. Indian Pines dataset. (a) Pseudocolor image for HSI. (b) Ground-truth map. (c) Color coding for each class.

TABLE I

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE INDIAN PINES DATASET

Class		Number of Samples		
No	Name	Training	Validation	Testing
1	Alfalfa	5	6	35
2	CornN	99	199	1130
3	CornM	58	116	656
4	Corn	16	33	188
5	GrassM	33	67	383
6	GrassT	51	102	577
7	GrassP	3	4	21
8	HayW	33	67	378
9	Oats	2	3	15
10	SoybeanN	67	136	769
11	SoybeanM	170	342	1943
12	SoybeanC	41	83	469
13	Wheat	14	29	162
14	Woods	88	176	1001
15	Buildings	27	54	305
16	Stone	9	12	72
	Total	716	1429	8104

TABLE II

NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE PAVIA UNIVERSITY DATASET

Class		Number of Samples		
No	Name	Training	Validation	Testing
1	Asphalt	198	965	5468
2	Meadows	559	2713	15377
3	Gravel	146	293	1660
4	Trees	153	437	2474
5	Metal sheets	40	196	1109
6	Bare soil	150	732	4147
7	Bitumen	66	189	1075
8	Bricks	110	536	3036
9	Shadows	47	135	765
	Total	1469	6196	35111

3) Salinas Dataset: The Salinas Dataset was captured by the AVIRIS hyperspectral sensor over the Salinas Valley of California, and its image size is 512 × 217. The dataset spectrum contains 224 bands, and after omitting the bands absorbed by atmospheric water, the remaining 204 bands are retained.

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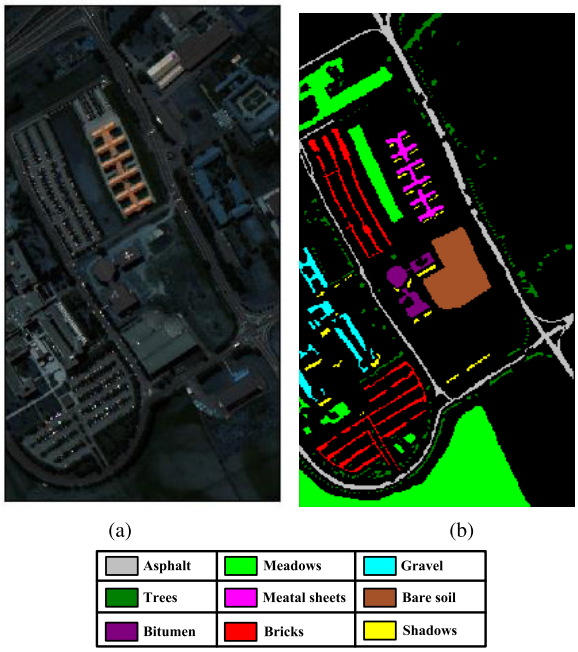


Fig. 7. Pavia University dataset. (a) Pseudocolor image for HSI. (b) Ground-truth map. (c) Color coding for each class.

TABLE III
NUMBER OF TRAINING, VALIDATION, AND TESTING SAMPLES FOR THE SALINAS DATASET

Class		Number of Samples		
No	Name	Training	Validation	Testing
1	Alfalfa	47	294	1668
2	CornN	75	548	3103
3	CornM	40	290	1646
4	Corn	28	205	1161
5	GrassM	54	394	2230
6	GrassT	80	582	3297
7	GrassP	72	526	2981
8	HayW	226	1657	9388
9	Oats	125	912	5166
10	SoybeanN	66	482	2730
11	SoybeanM	22	157	889
12	SoybeanC	39	283	1605
13	Wheat	19	134	763
14	Woods	30	156	884
15	Buildings	146	1068	6054
16	Stone	37	265	1505
	Total	1106	7953	45070

The Salinas dataset has a total of 54 129 sample pixels and 16 test categories. Table III lists the number of samples in the training set, validation set, and testing set applied in the experiment, and Fig. 8 presents the corresponding HSI pseudocolor image and total sample ground-truth map for this dataset.

B. Experimental Setup

In this section, we conduct a series of experiments to analyze and determine the parameters that best benefit the performance of CASST, including the number of blocks in TE, the number of heads in self-attention, and the initial learning rate of the network. In addition, we also empirically set default settings for some parameters as follows: 1) the network training epochs are 150, and the network training and testing batch sizes are 64 and 1000, respectively; 2) the gradient optimization algorithm uses an SGD algorithm, where the momentum is 0.9, and the weight decay is 1e-4.

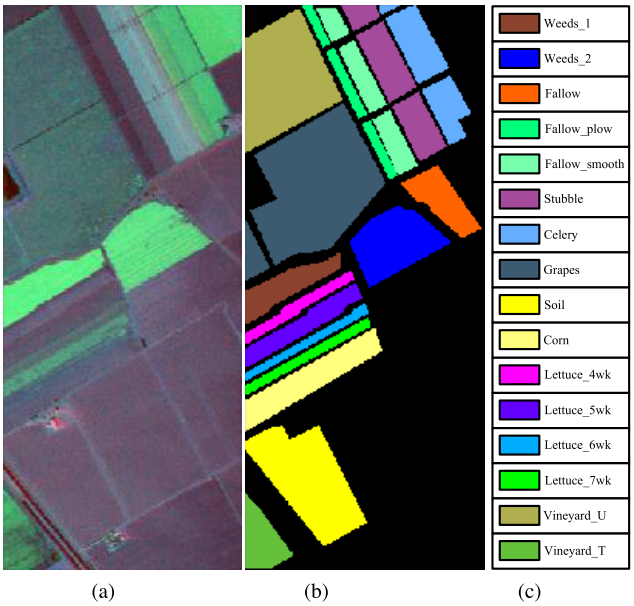


Fig. 8. Salinas dataset. (a) Pseudocolor image for HSI. (b) Ground-truth map. (c) Color coding for each class.

For the proposed CASST method, the number of Transformer blocks has a crucial impact on the final performance of the network. The deep TE structure can ensure more adequate extraction of spatial–spectral features. In the feature encoding process, the number of heads in the self-attention process also plays a key role, which determines whether the information acquisition of diverse feature subspaces is detailed. We carried out the mixed experiments on the number of heads and blocks in TE for three HSI datasets, and the results are shown in Fig. 9. The results show that when the number of blocks is two, the best classification results can be obtained on the three datasets. As the number of network layers deepens, the phenomenon of network degradation will occur. In addition, the optimal numbers of head parameters are eight, eight, and four for Indian Pines, Pavia University, and Salinas, respectively. In general, appropriately increasing the number of heads in self-attention can be more conducive to the information fusion of diverse feature subspaces, promote the complementarity of information, and enhance the identification of features. However, with the increase in the number of heads, the redundancy of information will also increase, which is not conducive to the improvement of performance.

The feature extraction process of the Transformer is different from that of CNN, which is more inclined toward the spatial information of the global context. To further analyze the impact of the sample spatial size on the pixel-wise HSI classification tasks, we analyze network performance comparisons when the sample spatial sizes range from 5 to 15, with a size growth interval of 2. Fig. 10 presents the experimental results of various spatial sizes on the three HSI datasets. Its shows that the optimal size of CASST on the Salinas dataset

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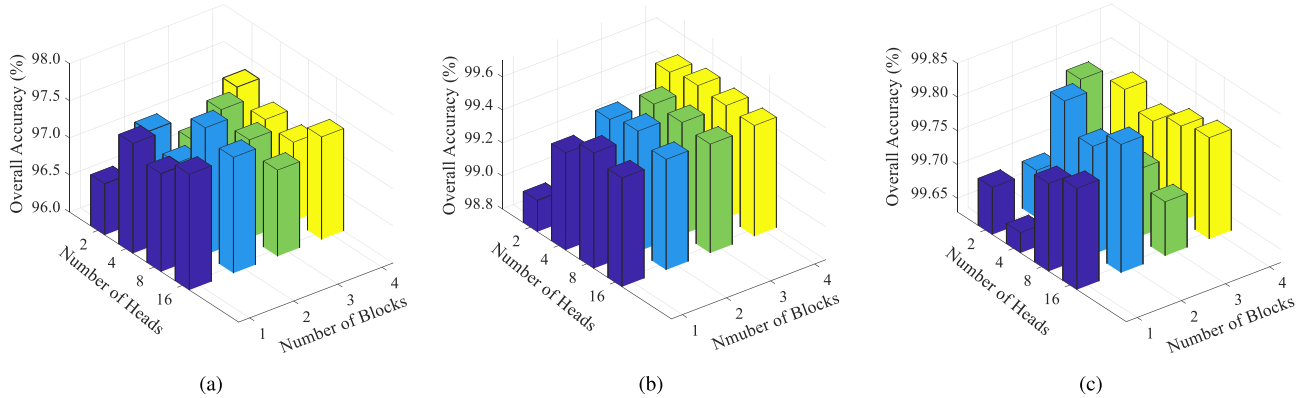


Fig. 9. Influence of the different numbers of heads and blocks on classification performance. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Salinas dataset.

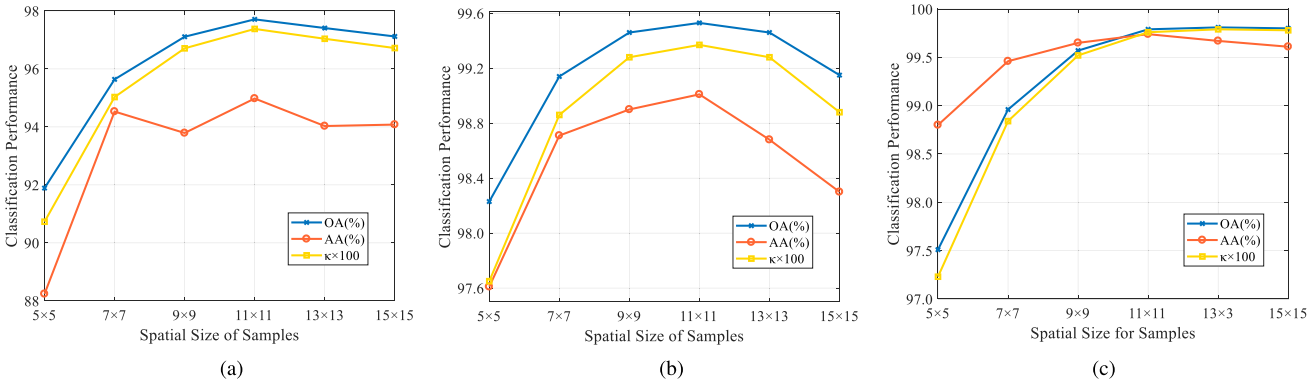


Fig. 10. Influence of the spatial size of samples on classification performance. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Salinas dataset.

TABLE IV
CLASSIFICATION PERFORMANCE OF DIFFERENT INITIAL NETWORK LEARNING RATES FOR THE PROPOSED METHOD ON THREE HSI DATASETS

Dataset	Metrics	Initial learning rate for training			
		0.05	0.01	0.005	0.001
IP	OA(%)	90.33	97.38	97.71	97.68
	AA(%)	83.86	94.24	94.98	94.77
	$\kappa \times 100$	88.97	97.02	97.38	97.35
PU	OA(%)	97.56	99.31	99.53	99.49
	AA(%)	95.85	98.70	99.01	98.92
	$\kappa \times 100$	96.76	99.08	99.37	99.32
SA	OA(%)	98.49	99.63	99.81	99.76
	AA(%)	98.60	99.70	99.76	99.63
	$\kappa \times 100$	98.32	99.59	99.79	99.73

is 13 and on the other two datasets is 11. It is worth noting that if the size of the sample cube is further increased, it will cause the interference of different categories of information on the sample category, and this phenomenon will be exacerbated by self-attention, resulting in information aliasing.

To minimize the impact of network training hyperparameters on the accuracy of CASST, a parameter analysis experiment is conducted on the initial learning rate during network training. For the selection of the initial learning rate, we set the values to {0.05, 0.01, 0.005, 0.001} and tested on three datasets. The experimental results show that when the initial learning rate is selected as 0.005, relatively better training results can be obtained on the three datasets.

C. Experimental Comparison With Competitive Methods

To verify the excellent performance of the proposed method, performance comparison experiments are conducted with various advanced hyperspectral classification algorithms on the Indian Pines dataset, Pavia University dataset, and Salinas dataset. Specifically, the traditional SVM classification algorithm [9] and the Super principal component analysis (PCA) algorithm [50] based on the spatial superpixel blocks are considered. For the field of deep learning, the two-dimensional CNN (2DCNN) [51], 3DCNN [52] and spectral–spatial residual network (SSRN) methods [53] based on the CNN model are compared. In addition, the recent emerging methods of ViT [45], SpectralFormer [41], and the spectral-spatial feature tokenization transformer (SSFTT) [42] based on the Transformer are also compared to prove the effectiveness of CASST.

1) SVM uses the SVM classifier to directly process the spectral features of HSI to realize the classification of ground objects. The code for SVM is implemented using the LIBSVM toolbox of MATLAB, and the model is trained with a Gaussian radial basis function (RBF) kernel function using fivefold cross-validation.
2) The SuperPCA [50] algorithm uses PCA in each superpixel block to achieve feature extraction in various homogeneous regions, effectively avoiding the confusion of spatial information. For the selection of parameters, the same parameter design is adopted for different datasets. Specifically, the number of PCs generated after PCA is 30, and the number of superpixel blocks is 100.

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TABLE V

CLASSIFICATION PERFORMANCE OF THE SVM, SUPERPCA, 2DCNN, 3DCNN, SSRN, VIT, SPECTRALFORMER, SSFTT, AND CASST CLASSIFICATION METHODS ON THE INDIAN PINES DATASET IN TERMS OF OA, AA, AND KAPPA

No	Class	Classification performance of various methods								
		SVM	SuperPCA	2DCNN	3DCNN	SSRN	ViT	SpectralFormer	SSFTT	CASST
1	Alfalfa	100.0	**100.0**	60.29	79.43	40.00	46.28	69.94	80.73	78.91
2	CornN	67.45	94.12	93.42	71.88	**99.20**	87.72	80.48	94.88	98.47
3	CornM	66.57	95.82	90.90	68.81	92.07	91.53	93.48	94.88	**97.64**
4	Corn	61.54	77.57	82.08	57.66	95.21	89.31	94.41	95.84	**96.54**
5	GrassM	89.31	**99.54**	94.12	75.46	95.82	94.10	86.08	93.27	90.77
6	GrassT	87.98	**99.71**	99.15	94.68	99.65	98.35	95.44	98.65	99.41
7	GrassP	85.71	**100.0**	83.81	68.10	47.62	88.57	85.19	94.40	96.07
8	HayW	87.50	**100.0**	99.79	96.77	100.0	100.0	75.10	99.91	100.0
9	Oats	**87.50**	38.30	66.67	56.00	00.00	44.00	79.43	70.00	80.00
10	SoybeanN	76.33	93.96	89.70	71.91	94.15	85.06	**99.38**	91.24	94.40
11	SoybeanM	75.25	93.80	95.14	81.91	97.12	91.92	93.89	98.35	**99.41**
12	SoybeanC	63.62	89.09	82.17	66.76	**98.29**	83.22	90.91	90.63	94.22
13	Wheat	96.70	**100.0**	99.57	85.25	97.53	99.69	95.55	98.17	99.90
14	Woods	90.46	**99.75**	98.36	97.69	99.70	98.99	72.79	99.05	99.33
15	Buildings	64.93	99.44	91.18	86.16	92.46	95.21	**100.0**	96.63	98.37
16	Stone	94.29	97.59	94.58	75.00	86.11	97.36	**100.0**	94.40	96.24
OA (%)		77.23(0.01)	95.22(0.01)	93.54(0.93)	80.38(0.94)	96.50(0.43)	91.93(0.29)	82.45(2.25)	96.13(0.62)	**97.71(0.11)**
AA (%)		80.95(0.01)	92.42(0.01)	88.81(1.75)	77.09(2.50)	83.43(2.31)	86.96(1.56)	88.19(1.14)	93.19(2.32)	**94.98(1.30)**
$\kappa \times 100$		73.85(0.01)	94.53(0.03)	92.62(1.07)	77.57(1.07)	96.00(0.49)	90.79(0.33)	80.03(2.45)	95.59(0.71)	**97.38(0.13)**

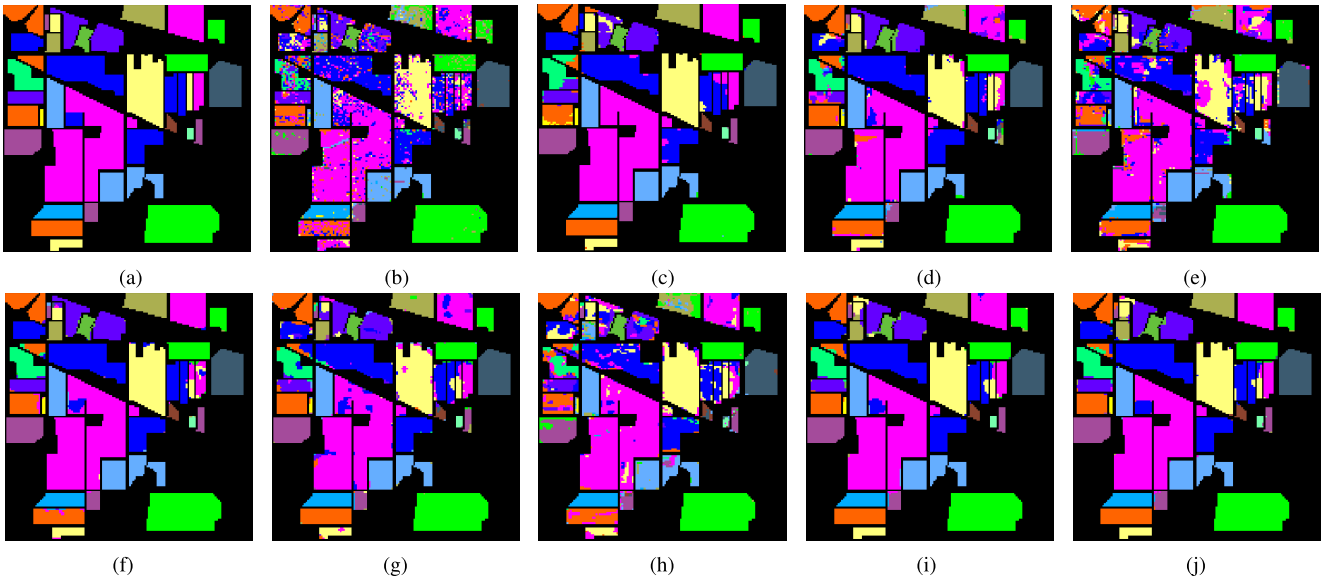


Fig. 11. Classification result maps for different comparison methods on the Indian Pines dataset. (a) Ground truth. (b) SVM (OA = 77.23%). (c) SuperPCA (OA = 95.22%). (d) 2DCNN (OA = 93.54%). (e) 3DCNN (OA = 80.38%). (f) SSRN (OA = 96.50%). (g) ViT (OA = 91.93%). (h) SpectralFormer (OA = 82.45%). (i) SSFTT (OA = 96.13%). (j) CASST (OA = 97.71%).

3) The 2DCNN structure uses three convolution kernels and includes batch normalization (BN), rectified linear unit (ReLU), and max pooling layers. A fully connected layer is used on the classification layer. The size of the network input space block is set to 9 × 9 on the Indian Pines dataset and 11 × 11 on the other two datasets.
4) The classification method of 3DCNN, in terms of structure design, is similar to that of 2DCNN, except for converting the 2-D process into 3-D. The rest of the settings are consistent with 2DCNN, and the same applies to the use of parameters.
5) The SSRN algorithm designs the neural network into a deeper level through the embedding of the residual structure, and it has extracted the spatial–spectral joint features with more semantic information. The parameter settings on the Indian Pines and Pavia University datasets are the default settings for the provided codes. For Salinas, the spatial block size is 9 × 9 and is trained for 200 epochs.
6) The hyperspectral classification algorithm based on ViT transplants the Transformer network from the field of NLP to the field of image classification. The size of the spatial input block is 9 × 9. During the image cutting process, the cutting size is set to nonoverlapping 3 × 3 to achieve the feature extraction process.
7) SpectralFormer is an application of another Transformer structure in the hyperspectral classification task.

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TABLE VI

CLASSIFICATION PERFORMANCE OF THE SVM, SUPERPCA, 2DCNN, 3DCNN, SSRN, VIT, SPECTRALFORMER, SSFTT, AND CASST CLASSIFICATION METHODS ON THE PAVIA UNIVERSITY DATASET IN TERMS OF OA, AA, AND KAPPA

No	Class	Classification performance of various methods								
		SVM	SuperPCA	2DCNN	3DCNN	SSRN	ViT	SpectralFormer	SSFTT	CASST
1	Asphalt	96.31	90.58	97.16	94.02	97.15	98.70	91.12	98.60	**99.62**
2	Meadows	94.68	95.55	99.83	98.55	99.96	99.88	99.42	**99.98**	99.96
3	Gravel	72.55	85.21	95.16	88.98	95.18	95.74	93.96	99.33	**99.36**
4	Trees	96.11	86.72	99.34	98.18	**99.76**	97.73	97.73	97.62	98.93
5	Metal sheets	99.54	97.38	99.83	98.87	100.0	100.0	**100.0**	98.90	99.90
6	Bare soil	90.98	95.11	98.80	86.28	99.59	97.28	81.06	99.83	**99.94**
7	Bitumen	72.99	95.55	93.65	83.20	**100.0**	99.51	89.32	99.73	99.96
8	Bricks	86.29	94.83	95.57	94.10	**99.24**	91.35	92.36	95.77	97.89
9	Shadows	**99.89**	92.84	99.62	99.42	99.87	97.94	99.33	94.01	95.54
OA (%)		92.07(0.00)	93.55(0.01)	98.47(0.16)	95.09(0.48)	99.18(0.22)	98.26(0.10)	94.68(1.05)	99.07(0.35)	**99.53(0.06)**
AA (%)		89.93(0.01)	92.64(0.01)	97.66(0.29)	93.51(0.64)	98.97(0.27)	97.57(0.12)	93.81(0.58)	98.24(0.68)	**99.01(0.10)**
$\kappa \times 100$		89.43(0.00)	91.40(0.03)	97.97(0.21)	93.46(0.65)	98.91(0.30)	97.68(0.13)	92.88(1.43)	98.77(0.46)	**99.37(0.08)**

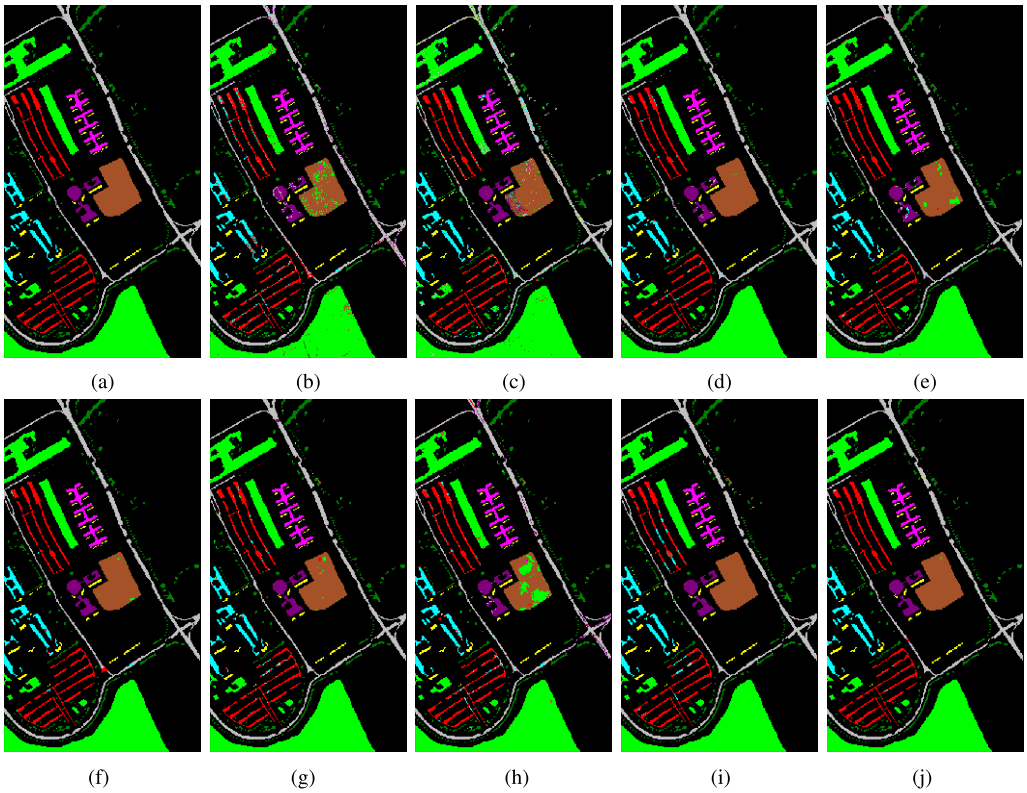


Fig. 12. Classification result maps for different comparison methods on the Pavia University dataset. (a) Ground truth. (b) SVM (OA = 92.07%). (c) SuperPCA (OA = 93.55%). (d) 2DCNN (OA = 98.47%). (e) 3DCNN (OA = 95.09%). (f) SSRN (OA = 99.18%). (g) ViT (OA = 98.26%). (h) SpectralFormer (OA = 94.68%). (i) SSFTT (OA = 99.07%). (j) CASST (OA = 99.53%).

A self-attention-based spatial–spectral feature extraction and classification process is implemented. The experimental parameters are designed using the default parameter settings of the provided code.

8) SSFTT is a spatial–spectral Transformer that combines the advantages of hybrid CNN and Transformer. The algorithm is implemented with the default parameters of the provided code.
9) The purpose of CASST is to effectively extract the discriminative spatial and spectral fusion features in HSI. The analysis and application of the parameters are proposed in part B of Section III. In terms of network training, it is the same as that of the three datasets. In all, 150 epochs were used.

Note that all the traditional methods are implemented on MATLAB, and all the methods based on deep learning are coded using Python 3.7.

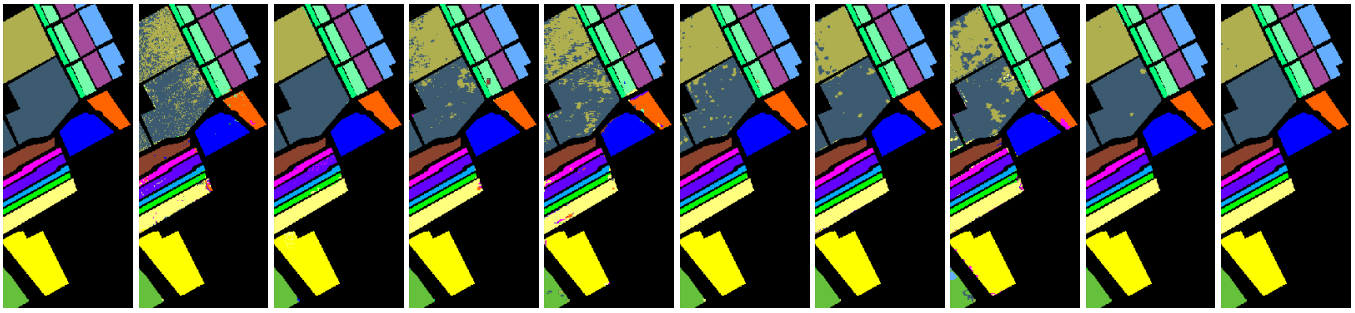
Tables V–VII summarize the classification performance (i.e., OA (%), AA (%), and Kappa) for various classification methods on the Indian Pines dataset, Pavia University dataset, and Salinas dataset. To ensure fairness of the experiment, each method uses the same number of training samples for training on the same device. After each experimental method is repeatedly tested ten times, the average value of each test result is taken as the final classification result. As the results presented in the tables indicate, the performances of CASST are sufficiently competitive compared with other methods on the three datasets.

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TABLE VII

CLASSIFICATION PERFORMANCE OF THE SVM, SUPERPCA, 2DCNN, 3DCNN, SSRN, VIT, SPECTRALFORMER, SSFTT, AND CASST CLASSIFICATION METHODS ON THE SALINAS DATASET IN TERMS OF OA, AA, AND KAPPA

No	Class	Classification performance of various methods								
		SVM	SuperPCA	2DCNN	3DCNN	SSRN	ViT	SpectralFormer	SSFTT	CASST
1	Weeds_1	99.66	100.0	99.86	99.96	99.97	99.69	97.10	99.75	**100.0**
2	Weeds_2	99.27	99.98	99.49	96.94	100.0	100.0	99.21	99.98	**100.0**
3	Fallow	93.07	98.66	99.54	88.48	99.99	99.99	77.84	99.95	**100.0**
4	Fallow_plow	97.61	95.31	99.22	97.45	99.67	**99.69**	98.50	98.06	95.45
5	Fallow_smooth	98.26	**99.68**	97.51	95.91	98.65	99.32	97.85	98.88	98.45
6	Stubble	100.0	99.99	100.0	99.71	100.0	**100.0**	99.69	99.72	98.39
7	Celery	99.60	99.98	99.86	98.28	99.97	99.92	99.49	99.65	**100.0**
8	Grapes	79.36	99.26	91.57	85.64	94.74	97.72	76.75	99.26	**99.93**
9	Soil	98.73	99.40	99.90	98.57	**100.0**	99.94	99.41	99.96	99.89
10	Corn	90.95	96.46	99.23	93.87	98.20	99.53	92.37	99.03	**100.0**
11	Lettuce 4wk	92.89	98.44	98.18	90.00	99.58	99.85	91.02	99.70	**100.0**
12	Lettuce 5wk	97.67	94.02	99.34	96.61	100.0	99.91	99.66	99.85	**100.0**
13	Lettuce 6wk	97.56	100.0	99.28	98.56	99.80	99.57	99.61	96.28	**100.0**
14	Lettuce 7wk	93.99	96.71	98.74	94.29	99.13	99.19	97.94	98.93	**99.77**
15	Vineyard_U	73.72	99.71	91.82	79.02	92.18	95.06	85.92	98.46	**99.72**
16	Vineyard_T	98.77	**100.0**	99.27	96.69	99.27	99.57	96.56	99.11	99.93
OA (%)		90.61(0.01)	99.00(0.01)	96.76(0.22)	92.06(0.30)	97.61(0.70)	98.73(0.26)	90.91(0.30)	99.29(0.11)	**99.81(0.04)**
AA (%)		94.44(0.01)	98.60(0.01)	98.30(0.14)	94.37(0.22)	98.82(0.33)	99.31(0.16)	94.31(0.24)	99.16(0.19)	**99.76(0.06)**
$\kappa \times 100$		89.54(0.01)	98.89(0.01)	96.39(0.25)	91.15(0.33)	97.34(0.78)	98.58(0.28)	90.24(0.32)	99.21(0.13)	**99.79(0.05)**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 13. Classification result maps for different comparison methods on the Salinas dataset. (a) Ground truth. (b) SVM (OA = 90.61%). (c) SuperPCA (OA = 99.00%). (d) 2DCNN (OA = 96.76%). (e) 3DCNN (OA = 92.06%). (f) SSRN (OA = 97.61%). (g) ViT (OA = 98.73%). (h) SpectralFormer (OA = 90.91%). (i) SSFTT (OA = 99.29%). (j) CASST (OA = 99.81%).

1) Indian Pines Dataset: Table V presents the performance comparison results of each classification algorithm on the Indian Pines dataset. The training, validation, and testing numbers of samples used are given in part A of Section III. It can be seen from the results that the traditional SVM only performs simple spectral feature processing, so it is easy to lead the phenomena of misclassification. With the patch-like distribution of each category in the Indian Pines dataset, the SuperPCA method harnesses the advantages of itself and achieves excellent classification accuracy, that is, 95.22% OA. The HSI classification algorithm for spatial feature extraction generally has superior performance. 2DCNN is a typical spatial feature extraction classification network with the basic framework of CNN, and 93.54% OA can be obtained on the Indian Pines dataset. The SSRN method, due to the combination of the structural characteristics of 2DCNN and 3DCNN, realizes the joint extraction process of detailed spatial–spectral features, and with the residual structure, the issue of network degradation is overcome, and the network is built in a deeper layer, which obtains 96.50% OA. Although the SSFTT method also realizes joint extraction of the spatial and spectral features by combining the advantage of CNN and Transformer, compared with the CASST proposed in this study, the design of the dual-branch spatial and spectral modeling process ensures more thorough information acquisition, and the SSCA model with weighted sharing intra Transformer block further realizes the fusion of diverse form features and a more detailed feature extraction process, which makes the proposed CASST method have better classification performance on the Indian Pines dataset.

2) Pavia University Dataset: As can be observed from the classification results presented in Table VI, for the Pavia University dataset with more sufficient training data, the CASST method can still have better classification performance and achieve the highest classification accuracy of 99.53% OA. Fig. 12 shows the full-label classification map of various methods. The visualization results obtained by the method of CASST have less misclassification and fewer noise pixels, especially compared with other methods that use sample cubes as input. In other words, the CASST method more effectively overcomes the issue of smooth edge transition, and its visual boundary contour is clearer and more detailed.

3) Salinas Dataset: Since the sample distributions of the Salinas dataset and the Indian Pines dataset are similar,

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TABLE VIII

CLASSIFICATION PERFORMANCE OF THE PROPOSED CASST METHOD WITH DIFFERENT ABLATION STRATEGIES ON THE THREE HSI DATASETS

Spatial branch	Spectral branch	SSCA	Indian Pines			Pavia University			Salinas		
			OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$	OA (%)	AA (%)	$\kappa \times 100$
×	√	√	89.88	85.49	88.42	96.06	94.30	94.73	97.34	96.07	97.04
√	×	√	97.24	93.02	96.85	99.31	98.71	99.08	99.71	99.61	99.68
√	√	×	97.37	93.77	97.00	99.45	98.92	99.27	99.74	99.67	99.72
√	√	√	97.71	94.98	97.38	99.53	99.01	99.37	99.82	99.70	99.80

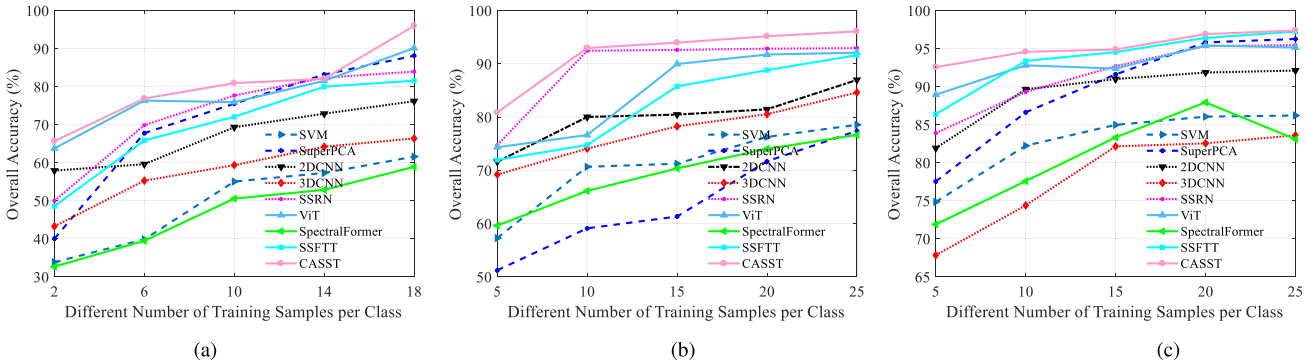


Fig. 14. Classification performance of various methods with different numbers of training samples on three datasets. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Salinas dataset.

i.e., they are both block-shaped, the performance comparison between the algorithms shows the same trend compared with the Indian Pines dataset. The CASST method proposed has a more stable spatial–spectral feature extraction process and realizes a classification process with strong robustness. Compared with other classification algorithms, it still obtained the highest classification accuracy of 99.81% OA. Fig. 13 also shows the full-label classification map of each classification method on the Salinas dataset. It can be observed that the CASST achieves a more accurate visualization process.

To further analyze the performance of CASST, the algorithm comparison experiments under different training samples are carried out to analyze the sensitivity of the CASST algorithm to sample changes. We randomly selected 5, 10, 15, 20, and 25 training samples for each class on the Pavia University and Salinas datasets, and 2, 6, 10, 14, and 18 samples on the Indian Pines dataset. As can be seen from the experimental results shown in Fig. 14, the classification accuracy of CASST changes steadily with the samples on the Pavia University and Salinas datasets, exhibiting its excellent sample sensitivity. Furthermore, CASST almost consistently outperforms the other contrasting methods under arbitrary training samples, which demonstrates the superiority of this method and its outstanding classification performance.

D. Ablation Experiments for the Proposed Method

The components to be ablated in the CASST method proposed in this study include spatial branch, spectral branch, and the SSCA module. By embedding each network module step by step, we conducted ablation analysis experiments on three datasets.

The specific results are summarized in Table VIII. When only the Transformer model is used for spectral sequence data processing, ideal classification accuracy cannot be obtained on any dataset. In general, the classification performance when the spatial feature is the main body of information is quite impressive. This shows that a single input of spectral sequence information is easily affected by noise pixels, resulting in misclassification of categories. In addition, it can be observed that the performance of the classification model is substantially enhanced on any dataset when the network can simultaneously implement spatial–spectral feature extraction, that is, when the spatial and spectral branches are embedded in the network at the same time. As a fusion module of information between different features, SSCA is better than the traditional linear feature-level fusion method at the end of TE and realizes the nonlinear fusion feature extraction based on the cross-attention intra Transformer block. It can be seen from the experimental results that a more reasonable and effective dual-branch feature fusion extraction method can further promote the full utilization of HSI data and information and help realize the classification process with strong robustness.

E. Analysis of Running Time

The time comparison analysis experiment compares the running time of CASST and other comparison methods, including training time and test time. In this experiment, all the time computing experiments are run on the same computer. In addition, the traditional algorithms, i.e., SVM and SuperPCA, are implemented and run on the MATLAB 2018a platform, while the rest of the deep-learning-based algorithm codes are run on Python 3.7. Section III Part A presents the dataset and the corresponding training and testing samples. The experimental results are presented in Table IX.

From the results in the table, it can be seen that the CASST method is not superior in terms of computational

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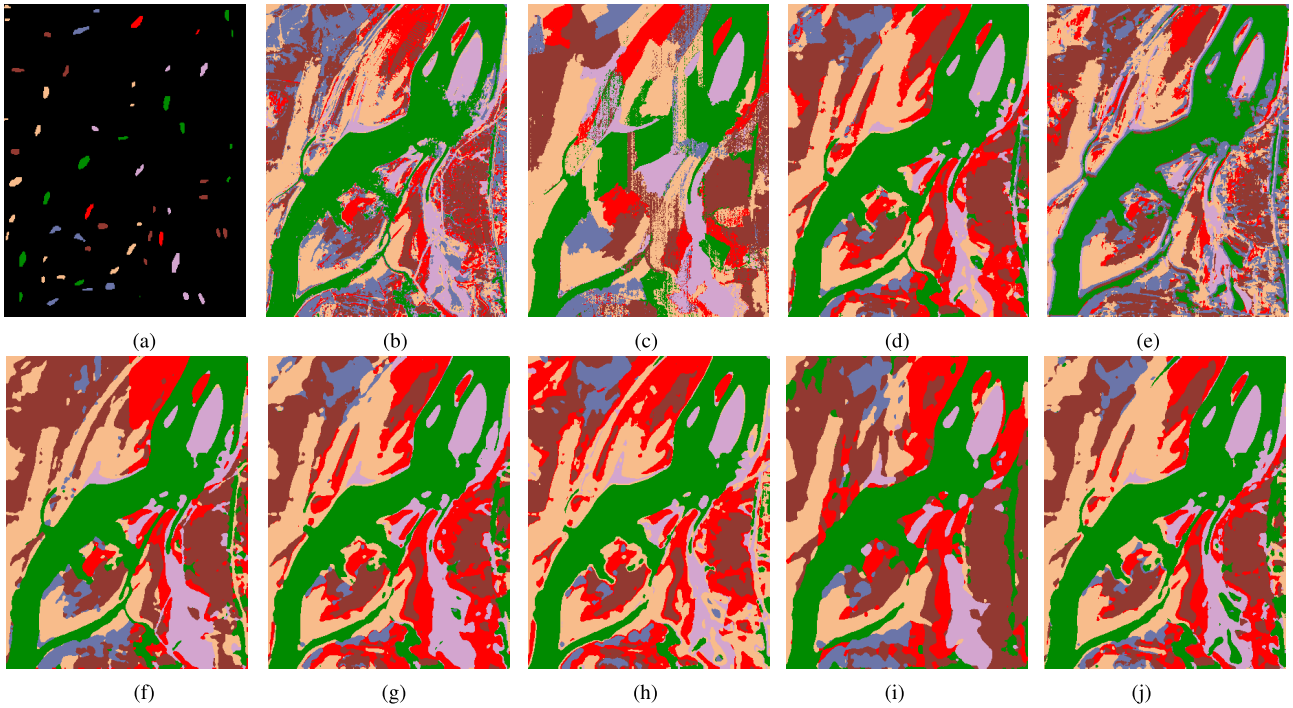


Fig. 15. Classification result maps for different comparison methods on the practical HSI dataset. (a) Ground truth. (b) SVM (OA = 95.51%). (c) SuperPCA (OA = 87.05%). (d) 2DCNN (OA = 92.64%). (e) 3DCNN (OA = 82.48%). (f) SSRN (OA = 98.33%). (g) ViT (OA = 99.60%). (h) SpectralFormer (OA = 98.89%). (i) SSFTT (OA = 98.69%). (j) CASST (OA = 99.82%).

TABLE IX
COMPARISON OF RUNNING TIME OF EACH METHOD ON DIFFERENT DATASETS

		Running time of various methods								
		SVM	SuperPCA	2DCNN	3DCNN	SSRN	ViT	SpectralFormer	SSFTT	CASST
IP	Train(s)	61.30	8.93	30.02	356.41	260.33	16.50	310.05	13.99	256.95
	Test(s)	2.94	0.49	3.76	10.18	2.83	1.87	6.76	0.54	4.88
PU	Train(s)	69.29	42.48	71.19	880.52	629.77	45.46	557.03	23.91	726.51
	Test(s)	12.97	9.04	4.88	24.13	10.03	4.56	8.29	2.38	18.35
SA	Train(s)	96.21	19.51	130.53	1885.52	782.52	43.95	1213.55	18.92	766.54
	Test(s)	11.26	3.24	9.08	57.94	15.91	5.73	19.25	3.11	23.25

TABLE X
TIME CONSUMPTION WITH OR WITHOUT WEIGHT SHARING ON THREE DATASETS

Dataset	Time (s)	Weighted sharing	
		√	×
IP	Training time	256.95	311.67
PU		726.51	840.34
SA		766.54	880.46

efficiency. The main reason is the complicated computing mechanism of the self-attention in the Transformer, resulting in high computational cost. It is worth noting that the spectral branch in the CASST method is more time-consuming than the spatial branch, and it is also one of the main time-consuming components in the network training process.

However, with the introduction of the spatial–spectral weighted sharing mechanism, the computational burden is relieved. Table X shows the comparison of computing time with and without weight sharing. It can be observed that the time consumption of the model after weight sharing is significantly reduced.

F. Analysis With an application HSI Dataset

To verify the effectiveness and excellence of the CASST method in practical applications, we conduct a comparative analysis experiment on the HSI data of an application scenario collected by the GaoFen-5 (GF-5) hyperspectral satellite. GF-5 is the first hyperspectral satellite in the world to realize comprehensive observation of the atmosphere and land. The dataset used in the experiment is one of the scenes in the HSI data of the Dongting Lake Basin acquired on January 22, 2019, with a spatial resolution of 30 m and a size of 456 × 352. After data preprocessing, 310 effective bands were retained, covering a range of 400–2500 nm. This scene has six categories, and 4816 ground-truth labels are contained.

During the experiment, 50 samples of each category are selected as training samples, 15% of the total samples as validation set samples, and the rest are used as the test set, to carry out the comparison experiments of CASST and various comparison methods. As a result, the CASST method has more performance advantages than other methods, and it obtains the highest classification accuracy of 99.82% OA. In addition, in the classification result map shown in Fig. 15,

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the CASST method avoids the generation of noise as much as possible and better handles the issue of spatial transition smoothing. Compared with other methods, the visualization map has more accurate classification performance and clearer class boundaries, which proves the superiority of CASST and its versatility in various classification scenarios.

IV. Conclusion and Future Lines

A spatial–spectral dual-branch sequence network for HSI classification based on the Transformer is proposed to overcome the disadvantage of incomplete feature utilization when applying the CNN model to classification tasks. The experiments on three real datasets, i.e., Indian Pines, Pavia University, Salinas, and an application scenario dataset compare several well-known traditional classification methods and state-of-the-art deep learning networks, demonstrating that the proposed CASST method can achieve better classification accuracy. We emphasize that the proposed dual-branch Transformer network has the ability to effectively capture deep spatial–spectral semantic feature information. Combined with the SSCA intra Transformer block and weighted sharing mechanism inter Transformer block, adaptive fusion extraction between various branches of features and utilization of information integrity are realized. In the future, research on the cooperative paradigm of global and local feature capture of the Transformer network will be our focus, and to further overcome the drawbacks of the Transformer structure, we will design a classification network with high performance and apply it to a small number of training sample classification scenarios.

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